

MACROECONOMIC ANNOUNCEMENT **SURPRISES UNDER FISCAL** **UNCERTAINTY**

Evidence from Fiscal Policy Uncertainty and U.S.
Government Shutdowns

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1. Introduction

A substantial body of research examines the effect of macroeconomic releases on financial markets. Early literature established that macroeconomic announcement surprises generate significant market responses (Hardouvelis, 1987; McQueen & Roley, 1993). While these early studies focused largely on daily returns, more recent work is high-frequency oriented (Andersen et al., 2003). Attention has since shifted toward the state-dependence of announcement reactions — the recognition that the same surprise can move markets differently depending on prevailing conditions, most notably uncertainty. Several papers study announcement reactions under specific types of uncertainty, including monetary policy uncertainty (Kurov & Stan, 2018) and economic uncertainty (Born et al., 2023).¹ Collectively, these studies indicate that the prevailing uncertainty regime conditions how markets react to macroeconomic news.

Yet the effect of *fiscal* uncertainty on the market's reaction to macroeconomic news has not been studied. This omission is striking, because Baker et al. (2016) find that fiscal policy is the single most important driver of overall economic policy uncertainty, rising sharply during budget disputes, debt-ceiling standoffs, and government shutdowns. The clearest illustration is the October 2013 shutdown: for sixteen days, federal statistical agencies suspended operations, delaying or cancelling scheduled releases, even as markets continued to price the macroeconomic data that did arrive amid acute fiscal stress. Whether — and how — such episodes alter the way equity markets process macroeconomic surprises is the question this thesis addresses.

This study examines whether fiscal uncertainty conditions the intraday reaction of S&P 500 futures to macroeconomic announcement surprises, using two complementary identification strategies: a continuous measure of fiscal policy uncertainty built from the fiscal-policy component of the Baker et al. (2016) index, and discrete U.S. government shutdown episodes. The analysis shows that fiscal uncertainty does not scale announcement reactions uniformly but reshapes them selectively: it activates equity reactions to several labour-market and real-activity releases that are otherwise insignificant, amplifies the reaction to a few others, dampens the bond market's response to real-activity surprises, and — for a subset of releases, most clearly Retail Sales — drives equity and Treasury markets in opposite directions, a pattern consistent with flight-to-safety. The remainder of this thesis is structured as follows. Chapter 2 reviews the literature on macroeconomic announcements, surprises, and uncertainty, and identifies the research gap. Chapter 3 develops the data and methodology, describing the futures data, the construction of the fiscal stress measures, and the regression specifications. Chapter 4 presents the empirical results for both the fiscal policy uncertainty measure and the government shutdown episodes. Chapter 5 concludes.

2. Literature study

2.1 Macroeconomic announcements and asset prices

A substantial body of empirical literature establishes that scheduled macroeconomic announcements constitute the primary drivers of asset price movements across markets. Fleming and Remolona (1997) provide early and compelling evidence in the bond market, showing that the 25 largest price movements in US Treasury markets from August 1993 to August 1994 were

almost exclusively associated with just-released economic announcements. While Fleming and Remolona's sample covers a single year of Treasury market data (August 1993 – August 1994), their core finding has since been confirmed over substantially longer samples and across multiple asset classes, including equity futures, as documented by Lahaye et al. (2011).

This pattern extends across asset classes. In foreign exchange markets, Andersen et al. (2003) document that US macroeconomic indicators produce significant and immediate price reactions across all major dollar exchange rates, characterized by a sharp jump at announcement time followed by little subsequent drift — consistent with efficient price discovery. In equity markets, Flannery and Protopapadakis (2002) show that certain macroeconomic announcements, particularly inflation and monetary variables, significantly affect daily aggregate stock returns, while others primarily influence return volatility — establishing a broad link between macro releases and equity pricing. More recently, Lahaye et al. (2011) confirm these findings using high-frequency intraday data across equity futures, bond futures and exchange rates, finding that 27-32% of equity futures jumps coincide with one of the 25 macro releases in their dataset — a figure far exceeding what would be expected under independence.

Importantly, the strength of this relationship is not uniform. McQueen and Roley (1993) demonstrate that conditioning on the state of the business cycle significantly strengthens the estimated news-return relationship in equity markets, with the same announcement producing meaningfully different reactions depending on whether the economy is in expansion or contraction. This early evidence of state-dependence in announcement pricing directly motivates the present study's focus on fiscal uncertainty regimes as a conditioning variable.

2.2 The role of announcement surprises

Multiple studies find that financial markets do not react to macroeconomic announcements per se, but rather to the surprise component — the deviation of the released value from market expectations. This distinction has important methodological implications: studies that rely solely on announcement indicators risk underestimating the true breadth and magnitude of macroeconomic news effects on asset prices.

Balduzzi et al. (2001) provide foundational evidence in the U.S. Treasury market, finding that the surprise component of 17 out of 23 economic announcements has a significant effect on bond prices, with employment and GDP releases generating the largest reactions. Importantly, Fleming and Remolona (1997) show that accounting for the magnitude and direction of the surprise renders seven additional announcements statistically significant beyond those identified using announcement indicators alone — reinforcing the empirical superiority of the surprise framework. Andersen et al. (2003) extend this evidence to foreign exchange markets, documenting that exchange rates react to macroeconomic surprises almost instantaneously, and introduce the standardized surprise measure — the deviation from consensus forecast scaled by its historical standard deviation — that has since become the methodological standard in the literature and is adopted in the present study.

In equity markets, Pearce and Roley (1985) establish the foundational principle that only the unanticipated component of macroeconomic announcements moves stock prices, with the anticipated component already reflected in prices prior to the release. However, their study finds only limited evidence of significant stock market reactions to macroeconomic surprises. McQueen and Roley (1993) later offer an explanation for this: the surprise-reaction relationship is time-varying, and by not conditioning on the state of the economy, earlier studies pool observations from expansions and contractions together, causing opposing reactions to partially offset each

other and weaken the estimated effect. Flannery and Protopapadakis (2002) confirm and extend the surprise framework more broadly, showing that several macroeconomic variables including CPI, PPI and employment have a statistically significant impact on aggregate stock returns through their surprise component.

However, the magnitude of the surprise-reaction relationship does not appear to be uniform across time and economic conditions. McQueen and Roley (1993) demonstrate that the stock market's response to macroeconomic news varies significantly with the state of the economy — a positive surprise in real activity is associated with higher stock prices in a weak economy but lower stock prices when the economy is already strong. Boyd et al. (2005) provide further evidence of this state-dependence, finding that the S&P 500 responds positively to bad employment news during expansions — when the interest rate channel dominates — but negatively during contractions, when growth expectation revisions take precedence. Together, these studies establish that the surprise-reaction relationship is itself state-dependent, a finding that serves as one of the theoretical foundations of the present study. As this foundation is crucial for the present study, it will be tested in hypothesis one (H1).

Fleming and Remolona (1997) offer a final and crucial observation: the magnitude of the market's reaction to a surprise is positively related to the degree of pre-announcement uncertainty. When uncertainty about the forthcoming number is high, the surprise carries greater informational value and markets react more strongly. This observation — that uncertainty about a single data release amplifies announcement reactions — naturally raises a broader question: if pre-release uncertainty shapes how markets price incoming surprises, does a more pervasive regime of fiscal uncertainty, such as that created by U.S. government shutdowns, systematically alter how equity markets process macroeconomic news? The following section examines the literature on uncertainty and asset prices.

2.3 Uncertainty

Over the last decades, uncertainty has become an increasingly important concept in the asset pricing literature. Veronesi (1999) provides an early theoretical foundation for the role of uncertainty in asset pricing. Using a rational expectations equilibrium model with learning, the paper shows that uncertainty about economic fundamentals affects stock prices and generates time-varying return volatility. When uncertainty is high, investors face greater difficulty in assessing the true state of the economy, which leads to increased sensitivity to new information and larger fluctuations in asset prices.

Anderson et al. (2009) find that uncertainty—proxied by the disagreement of professional forecasters about macroeconomic variables and used to construct implied expected market returns—appears to have greater explanatory power for expected market returns than risk, measured by return volatility. Uncertainty refers to situations in which both the outcome and the underlying probability distribution are unknown, whereas risk refers to situations where the distribution is known but outcomes are uncertain. Periods of elevated uncertainty are followed by higher expected excess market returns in the subsequent quarter, consistent with the broader implication in Pástor and Veronesi (2012) that uncertainty commands a risk premium. This finding also extends to the cross-section, where the price of uncertainty is significantly positive, implying that assets with greater exposure to uncertainty earn systematically higher returns across different assets and portfolios. While this evidence pertains to uncertainty in a broad sense, recent literature has shifted toward more granular measures—particularly economic and policy uncertainty—and their distinct implications for asset pricing.

Building on the general role of uncertainty in asset pricing, a substantial body of literature has focused more specifically on economic and policy-related uncertainty and its implications for financial markets. While early contributions emphasize uncertainty in a broad sense, more recent research highlights that uncertainty linked to government actions, policy decisions, and macroeconomic conditions plays a particularly important role in shaping asset prices and market dynamics. A key theoretical contribution in this literature is provided by Pástor and Veronesi (2012), who develop a rational expectations equilibrium model in which investors face uncertainty about future government policies. In their framework, policy changes are not fully predictable, and investors must form expectations about both the likelihood and impact of future policy actions. This uncertainty affects asset prices through time-varying discount rates: when policy uncertainty is high, investors demand higher compensation for bearing risk, leading to higher expected returns and lower contemporaneous stock prices. In addition, the model predicts increased stock market volatility, particularly around policy-related events, as investors update their beliefs about future policy outcomes. Importantly, the paper highlights that policy uncertainty is not merely a background condition but a priced risk factor that systematically influences financial markets. Complementing this theoretical perspective, Baker et al. (2016) provide an empirical framework to quantify economic policy uncertainty. They construct the widely used Economic Policy Uncertainty (EPU) index, which combines information from newspaper coverage, tax code provisions, and the dispersion of professional forecasts. This index allows for a systematic analysis of how policy uncertainty evolves over time and how it relates to economic and financial outcomes. Their findings show that policy uncertainty rises sharply during major political and fiscal events, including elections, debt ceiling disputes, and government shutdown threats. These periods of elevated uncertainty are associated with increased stock market volatility, reduced investment, and declines in economic activity. The empirical evidence thus reinforces the theoretical predictions by demonstrating that policy uncertainty has real and measurable effects on both financial markets and the broader economy. Overall, these studies establish that economic and policy uncertainty plays a central role in asset pricing by affecting both expected returns and market volatility. They also suggest that fiscal policy-related events, such as government budget disputes and shutdown risks, are important drivers of policy uncertainty and can generate substantial fluctuations in financial markets. This observation is particularly relevant in the context of the United States, where episodes of fiscal stress—most notably government shutdowns—are associated with heightened uncertainty about fiscal policy and economic conditions. While the existing literature provides strong evidence that policy uncertainty affects asset prices and economic activity, it primarily focuses on the direct impact of uncertainty on markets, such as its effect on volatility and risk premia. Less attention has been paid to how such uncertainty influences the way financial markets process new information. In particular, it remains unclear whether periods of elevated fiscal uncertainty alter the market reaction to macroeconomic announcement surprises, which constitute key information events for investors. The next section therefore focuses more specifically on fiscal uncertainty, with particular attention to periods of U.S. government shutdowns.

Among the components of economic policy uncertainty, fiscal policy emerges as the dominant driver. Baker et al. (2016) constructed the EPU index, decomposing policy uncertainty across several categories including fiscal policy, monetary policy, and a range of others. They found that fiscal policy uncertainty is, by a considerable margin, the most important driver of overall policy uncertainty, a pattern that became substantially more pronounced in the most recent years of their sample, namely 2010–2013. This finding is corroborated by a later study of Baker et

al. (2021), who constructed an Equity Market Volatility (EMV) tracker based on newspaper coverage. Among all policy categories tracked in the EMV index, fiscal policy articles accounted for the largest share — 35% — suggesting that fiscal policy constitutes the most prominent policy-driven source of equity market volatility. This motivates treating fiscal uncertainty as a distinct and economically meaningful regime, rather than a subcategory of broader policy uncertainty.

Croce et al. (2012) provide formal theoretical grounding for treating fiscal uncertainty as a distinct asset pricing concept. In their endogenous growth model with robust preferences, agents who are uncertain about the future path of tax distortions demand a compensation that the authors formally define as the market price of fiscal uncertainty — a component of the stochastic discount factor that is empirically distinct from the ordinary market price of risk. This distinction is economically meaningful: unlike standard consumption risk, fiscal uncertainty arises from the long-run, persistent nature of tax distortions, which are inherently difficult to detect and hedge against. Their calibration demonstrates that even modest levels of fiscal uncertainty produce substantial welfare losses and depress long-run growth by discouraging investment in innovation. Collectively, these results establish that fiscal uncertainty is not merely a subcategory of broader economic uncertainty, but a theoretically grounded and economically consequential concept with its own distinct asset pricing implications.

Despite these measurement challenges, the EPU index provides a valuable systematic tool for identifying periods of elevated fiscal uncertainty at the monthly level. Its fiscal policy component, which tracks newspaper coverage of taxes and government spending, allows for a structured comparison of announcement reactions across fiscal stress and non-stress regimes over a long sample period — an advantage that discrete event-based measures alone cannot provide. The specific construction of the fiscal stress measure employed in this study is discussed in the data and methodology section.

Measuring fiscal uncertainty empirically is non-trivial. Broad uncertainty indices such as the EPU or VIX capture many overlapping sources of uncertainty and respond to the very market conditions under study, making it difficult to isolate fiscal stress specifically. Importantly, Baker et al. (2016) themselves acknowledge that the 1995-1996 government shutdowns generated little EPU elevation despite extensive press coverage — suggesting that media-based indices may fail to capture objective periods of fiscal dysfunction entirely. US government shutdowns, by contrast, represent discrete, precisely dated episodes of fiscal stress triggered by Congressional budget impasses rather than market developments, making them a more direct and consistent identifier of fiscal uncertainty regimes. The construction and validation of this measure is discussed in detail in the data and methodology section.

As established in the preceding sections, financial markets react differently to macroeconomic news depending on the state of the economy — as demonstrated by Boyd et al. (2005) and McQueen and Roley (1993) — suggesting that the announcement reaction is inherently state-dependent. A growing body of empirical literature extends this insight to uncertainty regimes specifically, demonstrating that the prevailing type of uncertainty systematically conditions how markets process macroeconomic surprises. Crucially, the direction and magnitude of this conditioning effect depends on the nature of the uncertainty regime and the channel through which it operates — a finding that has direct implications for the present study.

Kurov and Stan (2018) examine whether monetary policy uncertainty influences the reaction of stock, bond, foreign exchange, and crude oil markets to 33 U.S. macroeconomic announcements using intraday futures data. They distinguish between a direct effect, driven by the informational content of the release, and an indirect effect operating through monetary policy

expectations. For equity markets, higher monetary policy uncertainty weakens the announcement reaction through the policy anticipation channel: good economic news increasingly signals imminent Fed tightening, raising discount rates and counteracting the positive direct effect on stock prices. Their findings raise the possibility that state-dependence in announcement reactions documented in prior studies has more to do with monetary policy expectations than with the business cycle *per se*.

Born et al. (2023) extend the analysis to economic uncertainty more broadly, examining how expectation dispersion and aggregate economic uncertainty condition the S&P 500's reaction to six major macroeconomic indicators across 1,671 releases. Higher expectation dispersion attenuates the market reaction, while higher economic uncertainty amplifies it — rationalized through an informational value channel whereby incoming data becomes more valuable to investors when fundamental uncertainty is elevated. Crucially, Born et al. (2023) also confirm that when monetary policy uncertainty is used as the conditioning variable, the amplification effect reverses and the equity reaction weakens, directly corroborating Kurov and Stan (2018) and reinforcing that the direction of the effect is determined by the nature of the uncertainty regime rather than by uncertainty *per se*. This informational value channel is directly relevant to the fiscal uncertainty context: if elevated fiscal stress increases fundamental uncertainty about economic outcomes, incoming macroeconomic data may carry greater informational value to investors, generating amplified announcement reactions — a hypothesis that the present study tests empirically using the fiscal policy EPU sub-index as a conditioning variable

Benchimol et al. (2023) complement this evidence by examining how financial uncertainty conditions the stock market's reaction to monetary policy surprises. Using high-frequency data on 192 FOMC announcements from 1994 to 2016, they show that equity markets respond more aggressively to monetary policy surprises during periods of high financial uncertainty, with the amplifying effect being stronger for accommodative than restrictive surprises. These results hold outside recession and bear market periods, suggesting the phenomenon is not confined to crisis episodes.

Bird and Yeung (2012) extend the evidence beyond macroeconomic releases and the US context, examining earnings announcement surprises in the Australian stock market. Under high uncertainty, the symmetry between positive and negative surprise reactions breaks down — bad news generates a significantly stronger negative reaction while good news is largely dismissed, consistent with investors adopting a worst-case scenario framework under uncertainty.

Taken together, these studies establish that the prevailing uncertainty regime systematically conditions how markets process announcement surprises, with the direction and magnitude of the effect depending on the type of uncertainty and the channel through which it operates. This heterogeneity has a direct implication for the present study. The dominant channels identified in the existing literature — the policy anticipation channel, through which monetary policy uncertainty dampens equity reactions, and the informational value channel, through which economic uncertainty amplifies them — are both plausibly absent during US government shutdowns, which are driven by Congressional budget impasses rather than monetary policy expectations or fundamental economic signals. Whether fiscal uncertainty amplifies or dampens how equity markets react to macroeconomic announcement surprises therefore cannot be inferred from existing evidence. The present study addresses this gap directly, employing two complementary identification strategies — the fiscal policy EPU sub-index as a continuous measure of fiscal stress, and government shutdowns as discrete, precisely dated episodes of acute

fiscal dysfunction — to examine whether and how fiscal uncertainty regimes alter the surprise-return relationship in the S&P 500. This effect will be studied in H2 and H2a

2.4 Research gap and contribution

The existing literature establishes several key insights regarding the relationship between macroeconomic information and asset prices. First, a large body of evidence shows that financial markets react strongly to scheduled macroeconomic announcements, with price adjustments occurring rapidly at the time of release. Second, it is well documented that these reactions are driven not by the announcements themselves, but by their surprise component — the deviation from prior market expectations. Third, the magnitude and even direction of these reactions are not constant, but depend on prevailing economic conditions, suggesting that announcement pricing is inherently state-dependent. More recent literature extends this state-dependence to uncertainty regimes, demonstrating that different types of uncertainty condition announcement reactions through different channels and can produce opposite effects: while economic uncertainty tends to amplify market reactions through an informational value channel, monetary policy uncertainty may dampen them through a policy anticipation channel. These findings imply that the impact of macroeconomic news depends critically on the nature of the uncertainty regime.

Yet a notable gap remains. Existing studies focus primarily on general economic uncertainty, financial uncertainty, or monetary policy uncertainty, while largely neglecting fiscal uncertainty as a distinct conditioning regime. This neglect is particularly striking given that the literature on policy uncertainty consistently identifies fiscal policy as by far the most important driver of overall economic policy uncertainty, rising sharply during periods of political and budgetary stress. No existing study, to the best of my knowledge, examines whether this empirically dominant form of policy uncertainty systematically alters how equity markets respond to macroeconomic announcement surprises. The studies closest to this question either examine uncertainty in general terms, focus on monetary or financial uncertainty specifically, or study the level effects of fiscal uncertainty on asset prices without considering its dynamic interaction with incoming information. Notably, the need for this type of investigation was anticipated early in the literature. Andersen et al. (2003), in one of the most influential studies of announcement effects in foreign exchange markets, explicitly called for future research exploiting the broader implications of Veronesi (1999) — namely that news effects are in general a function of state uncertainty — by interacting announcement surprises with measures of state uncertainty. The present study responds to this call in the equity market context, using fiscal uncertainty regimes as the conditioning state variable.

This study employs two complementary identification strategies to examine whether fiscal uncertainty conditions the intraday surprise-return relationship in the S&P 500. The primary identification uses the fiscal policy categorical sub-index of the EPU index — which tracks newspaper coverage of taxes and government spending — to classify months of elevated fiscal stress, providing a continuous measure that captures the broad climate of fiscal policy uncertainty across the full sample period. As a media-based measure, the fiscal policy EPU sub-index also captures the fiscal uncertainty that is actually transmitted to market participants through the information environment — making it conceptually well-suited to studying how uncertainty shapes market behavior, since asset prices respond to perceived rather than abstract institutional uncertainty. Government shutdowns serve as a second, sharper identification strategy: as discrete, precisely dated episodes of actual fiscal dysfunction triggered by Congressional budget impasses,

they complement the EPU-based results by capturing acute fiscal crises that may not be fully reflected in newspaper-based indices. Government shutdowns offer several advantages over existing uncertainty proxies. Unlike the EPU index — which Baker et al. (2016) themselves acknowledge failed to register the 1995–1996 shutdowns despite their political significance — shutdown dates provide an objective, complete, and consistently defined record of fiscal dysfunction that does not depend on media framing or market conditions. Unlike VIX, they are not endogenous to the equity market dynamics under study. Moreover, Anderson et al. (2009) explicitly distinguish between risk — measured by return volatility — and uncertainty — measured by forecaster disagreement — demonstrating that the two carry separate and distinct asset pricing premiums. This conceptual separation implies that volatility-based proxies such as VIX, which capture the risk dimension, may not adequately reflect the uncertainty dimension that matters for announcement reactions in the Knightian sense. Government shutdowns, by directly identifying a regime of objective fiscal dysfunction, sidestep this conceptual conflation. Unlike continuous uncertainty measures, they also define discrete, precisely dated regimes that allow for a clearer comparison of announcement reactions across fiscal stress and non-stress periods.

The contribution of this study is therefore threefold. First, to the best of my knowledge, it provides one of the first systematic examinations of whether fiscal uncertainty conditions the equity market's reaction to macroeconomic announcement surprises, employing dual identification through both a continuous EPU-based fiscal stress measure and discrete government shutdown episodes — extending the uncertainty-announcement reaction literature to a regime that has not previously been examined. Second, it contributes to the literature on government shutdowns and financial markets by moving beyond the study of shutdown effects on price levels and volatility to examine their interaction with the information-processing mechanism. Third, it speaks to the broader question of whether the channels through which uncertainty affects announcement reactions — policy anticipation and informational value — are the only relevant mechanisms, or whether fiscal dysfunction creates a distinct conditioning environment that operates through a different channel entirely. In sum, this study seeks to bridge the gap between the literature on macroeconomic announcement surprises and the literature on policy uncertainty by analysing how fiscal uncertainty regimes shape the processing of macroeconomic information in equity markets.

2.5 Hypotheses

The literature reviewed above shows that macroeconomic announcement surprises affect financial markets, but that the magnitude and direction of these effects may depend on the prevailing uncertainty environment. Since this thesis focuses on fiscal policy uncertainty, the central question is whether markets react differently to macroeconomic announcement surprises when fiscal stress is elevated. The hypotheses below translate this research question into empirically testable predictions.

First, before examining fiscal stress regimes, it is necessary to establish that macroeconomic announcement surprises generate measurable intraday market reactions in the sample used in this study. Consistent with the macroeconomic announcement literature, unexpected information in macroeconomic releases is expected to be reflected rapidly in S&P 500 futures prices.

H1: Larger macroeconomic announcement surprises generate larger intraday market reactions in the S&P 500.

The main hypothesis concerns the role of fiscal policy uncertainty. Prior literature suggests that uncertainty can increase the informational value of new macroeconomic data, as investors rely more heavily on incoming information when the economic environment is harder to interpret. Therefore, during periods of elevated fiscal policy uncertainty, macroeconomic announcement surprises are expected to generate stronger equity market reactions.

H2: Fiscal policy uncertainty amplifies the sensitivity of equity market returns to macroeconomic announcement surprises.

However, this amplification effect is unlikely to be identical across all macroeconomic releases. Announcements differ in their economic content: inflation releases, activity indicators, trade data, and confidence measures may each affect equity prices through different channels. Therefore, the fiscal uncertainty effect is expected to vary across announcement types.

H2a: The amplification effect of fiscal policy uncertainty differs across announcement types.

In addition to the fiscal policy uncertainty index, this thesis uses U.S. government shutdowns as a second independent identification strategy. Shutdowns represent discrete, precisely dated episodes of acute fiscal stress triggered by Congressional budget impasses rather than market developments, allowing the fiscal uncertainty channel to be tested independently of the continuous FPU measure. The number of macroeconomic announcements occurring during active shutdown periods is limited, however, so while the effect during active shutdowns may be economically meaningful, it is expected to be statistically weaker than the main fiscal policy uncertainty results.

H3a: Government shutdown episodes are associated with altered announcement sensitivity during the active shutdown period, though the effect is expected to be weak given the limited number of active-shutdown observations.

Fiscal stress may also emerge before an official shutdown begins. In the period leading up to a shutdown, investors may anticipate political gridlock, delayed fiscal decisions, or disruptions to government operations. As a result, markets may already process macroeconomic announcements differently before the official start of the shutdown. Moreover, under heightened uncertainty about the fiscal outlook, each incoming macroeconomic release may carry greater informational value, as investors have fewer anchor points for forming expectations — consistent with Veronesi (1999), who shows that the price impact of news is amplified when investors are uncertain about the prevailing economic regime.

H3b: Markets anticipate fiscal stress prior to government shutdowns, resulting in amplified announcement sensitivity in the pre-shutdown window relative to non-episode periods.

Similarly, fiscal stress may persist after a shutdown officially ends. Even once government operations resume, uncertainty about fiscal policy, delayed data releases, and unresolved budgetary issues may continue to affect the way markets interpret macroeconomic information.

H3c: Fiscal stress persists beyond the official end of a government shutdown, resulting in amplified announcement sensitivity in the post-shutdown window relative to non-episode periods.

Finally, government shutdowns can delay the publication of certain macroeconomic indicators. When delayed announcements are eventually released, they may contain accumulated information that markets had been unable to observe during the shutdown. This information backlog may lead to stronger market reactions upon release.

H3d: Announcements that were delayed due to a government shutdown trigger stronger market reactions upon release than announcements released on schedule, consistent with an information backlog mechanism.

Beyond the equity market, fiscal uncertainty may also affect the bond market in ways that reveal the underlying mechanism. If elevated fiscal uncertainty leads investors to shift from equities to safe-haven assets, the same announcement surprise that dampens equity sensitivity may simultaneously amplify Treasury market sensitivity. Such opposing directional effects across markets would be consistent with flight-to-safety dynamics and would provide a market-level mechanism that a pure noise or attention story cannot easily explain.

H4: Under elevated fiscal policy uncertainty, equity and Treasury markets respond in opposite directions to the same announcement surprise for certain announcement types, consistent with flight-to-safety dynamics.

Together, these hypotheses test whether fiscal stress affects the pricing of macroeconomic information both through the continuous FPU measure and through discrete government shutdown episodes. The FPU-based hypotheses form the main empirical contribution, while the shutdown-based hypotheses provide independent event-based evidence on specific fiscal stress episodes.

3. Methodology

3.1 Data

3.1.1. S&P 500

The S&P 500 is proxied by E-mini S&P 500 futures (ticker: ES), obtained from Databento, which sources data directly from the Chicago Mercantile Exchange (CME). ES futures trade continuously except for a daily maintenance break from 17:00 to 18:00 ET, making them suitable for studying intraday price reactions to macroeconomic announcements released outside regular stock market trading hours. The majority of announcements in the sample are released at 8:30 ET, when the cash equity market is closed.

Both the ES series and the 10-year Treasury Note futures series (ticker: ZN), used for the cross-market analysis, are constructed as continuous front-month contracts (Databento continuous symbols ES.v.0 and ZN.v.0). Databento maps the continuous series to the actual front-month contract on each date using a volume-based roll rule, switching to the next contract once trading volume migrates to it. Following Databento's convention, the continuous prices are unadjusted:

rolls are not back-adjusted, so a level discontinuity occurs at each contract change. This is immaterial for the present study, however, as none of the contract rolls over the full sample period coincide with an announcement date — verified directly against Databento's contract mapping — so no measured announcement return spans a roll.

Data are obtained at one-minute frequency and aggregated to five-minute bars, providing a balance between granularity and noise reduction that is standard in the high-frequency macro-announcement literature. The data cover the period from June 7, 2010 to May 18, 2026. Bar timestamps mark the start of the five-minute interval, which was empirically verified by confirming that the price move associated with a known 8:30 release appeared within the 8:30 bar rather than the subsequent bar.

Returns are computed as log returns, defined as $\ln(P_{\text{after}} / P_{\text{before}})$, where P_{before} and P_{after} denote the closing prices of the bars immediately before and after the announcement window respectively. Log returns are standard in the finance literature for their time-additivity and approximate symmetry around zero. Expressing returns in basis points ensures that regression coefficients are interpretable in economically meaningful units — for example, a coefficient of 5 indicates that a one-standard-deviation surprise is associated with a five basis point return.

3.1.2. Macroeconomic announcements

This study examines equity market reactions to 16 scheduled macroeconomic announcements, selected to cover the main dimensions of US economic activity. The selection builds on McQueen and Roley (1993), four of whose significant announcements are retained: the Consumer Price Index (CPI), Industrial Production (IP), Unemployment Rate, and US International Trade Balance — the modern equivalent of the Merchandise Trade Balance used in their study. M1 money stock, which was significant in their study, is excluded because the series was redefined in May 2020 — savings deposits were reclassified as transaction accounts and folded into M1, producing a large structural break in the aggregate — which destroys the continuity required for a consistent surprise series over the sample period. Producer Price Index (PPI), although significant in their study, is not included due to incomplete data availability over the full sample period. Nonfarm Payrolls (NFP), while insignificant in McQueen and Roley's (1993) sample, is included because the present study covers a different sample period and because NFP is commonly included in more recent studies of market reactions to macroeconomic news, such as Kurov and Stan (2018).

Following Andersen et al. (2003), the announcements are organized into thematic categories: real activity, consumption, investment, government purchases, net exports, prices, and forward-looking indicators. Table 1 presents the full list of announcements along with their releasing agency, release time, and frequency. Beyond the McQueen and Roley (1993) announcements, the sample includes housing market indicators (New Home Sales and Existing Home Sales, both expressed as seasonally adjusted annual rates), consumer indicators (Consumer Confidence and Consumer Credit, measuring the monthly change in outstanding consumer credit), activity indicators (GDP advance estimate, Retail Sales, Durable Goods Orders, and Construction Spending), forward-looking survey indicators (the ISM Manufacturing PMI — a composite index of purchasing manager surveys with readings above 50 indicating expansion — and its services-sector counterpart, the ISM Non-Manufacturing PMI, which measures activity in non-manufacturing industries on the same diffusion-index basis), and the US Federal Budget Balance, which captures monthly federal government income and expenditure.

All announcements are released monthly except GDP, which is quarterly. All releases are seasonally adjusted except the Federal Budget Balance. Initial release values are used throughout, as these represent the figures that drive market reactions at the time of release; subsequent revisions are excluded. Observations from April through July 2020 are excluded for all announcement types due to extreme values driven by the COVID-19 pandemic. During this period, both announcement surprises and intraday returns exhibited outliers of an order of magnitude larger than the rest of the sample, reflecting the simultaneous collapse in economic activity, unprecedented forecast errors as consensus forecasters had no basis for predicting COVID-affected releases, and disruptions to normal statistical agency operations. Retaining these observations would distort surprise distributions and dominate regression estimates.

Consensus forecasts are obtained from the Reuters Poll via Refinitiv. Although Refinitiv exports include release timestamps, a small number of timestamps deviated from the official release schedule. Because vendor-provided timestamps may reflect transmission or processing delays rather than the official announcement time, official release times from the corresponding agencies are used throughout. LSEG developer-support discussions similarly note that timestamp-delay issues can arise from data transmission and client-side infrastructure conditions¹. A brief descriptive overview of the announcements is provided in Table 1.

For each announcement, the surprise is defined as the deviation of the released figure from the consensus forecast, scaled by the historical standard deviation of that announcement type's surprises:

$$S_{j,t} = \frac{(A_{j,t} - F_{j,t})}{\sigma_j}$$

¹See LSEG Developer Community, "quote delay problem".

TABLE 1 — MACROECONOMIC ANNOUNCEMENTS

Announcement	Source	Time	N	Mean ret (bps)
Real activity				
NFP	BLS	8:30	187	28.87
Unemployment rate	BLS	8:30	186	28.96
Retail sales	Census	8:30	186	9.66
Industrial production	FRB	9:15	187	4.29
Consumer credit	FRB	15:00	186	5.86
Consumption				
New home sales	Census	10:00	184	9.29
Existing home sales	NAR	10:00	188	8.73
Investment				
Durable goods	Census	8:30	184	7.34
Construction spending	Census	10:00	186	15
Output				
GDP	BEA	8:30	62	10.54
Government				
Federal budget	FMS	14:00	188	6.96
Net exports				
US International trade	BEA	8:30	188	8.45
Prices				
CPI	BLS	8:30	184	23.81
Forward-looking				
Consumer confidence	CB	10:00	187	9.68
ISM PMI	ISM	10:00	187	15.46
ISM N-Mfg PMI	ISM	10:00	183	13.17

Note: Mean |ret| is the mean absolute $[0, +5]$ minute ES return (announcement-bar open-to-close) in basis points. N = number of observations after COVID exclusion (Apr–Jul 2020).

3.1.3. Fiscal stress regimes

To identify periods of elevated fiscal uncertainty, this study draws on the categorical EPU data constructed by Baker et al. (2016). The headline US EPU index is a broad monthly measure based on scaled article-frequency counts across ten leading newspapers. Beyond it, Baker et al. construct a set of categorical EPU indices covering specific policy areas, including monetary policy, taxes, government spending, and sovereign debt or currency crises.

This study employs the fiscal policy sub-index, hereafter referred to as the Fiscal Policy Uncertainty (FPU) index, which combines the taxes and government spending categorical indices. To be included in the FPU index, an article must satisfy the baseline EPU criteria and additionally contain terms related to either taxes or government spending — such as 'tax', 'taxation', 'government spending', 'federal budget', 'budget battle', 'budget deficit', or 'federal debt'. As a media-based measure, the FPU index captures the fiscal uncertainty that is actually transmitted to market participants through the information environment, making it conceptually well-suited to studying how uncertainty shapes market behaviour, since asset prices respond to perceived rather than abstract institutional uncertainty.

The FPU index enters the regressions as a continuous variable, standardized over the estimation sample to have mean zero and unit standard deviation. Standardization makes the interaction coefficient directly interpretable: β_3 measures the change in the surprise-return sensitivity associated with a one-standard-deviation increase in fiscal policy uncertainty, facilitating comparison across announcement types. The continuous form is preferred over a threshold-based dummy as the primary measure of fiscal uncertainty, as it is less arbitrary and exploits the full variation in the index rather than collapsing it into a binary regime indicator; the dummy specification is reported as a robustness check. The timing with which the monthly index is matched to each announcement — contemporaneous versus lagged — is discussed below.

As a robustness check, a binary fiscal stress dummy is constructed at the monthly level by applying a threshold at the 75th percentile of the FPU distribution over the sample period from January 2010 to April 2026. This upper-quartile cutoff identifies periods of unusually elevated fiscal policy uncertainty relative to the estimation sample, while retaining a sufficient number of stressed observations for reliable estimation. A month is classified as a fiscal stress period if the FPU index exceeds this threshold, and as a non-stress period otherwise. Each announcement falling within a fiscal stress month is assigned a value of one on this dummy, and zero otherwise. Across the full sample, 676 of 2,855 announcement observations — approximately 25.1% — are classified as fiscal stress periods. The sensitivity of results to this threshold choice is examined in the robustness section, where the 70th, 80th, and 90th percentile thresholds are tested as alternatives.

A timing consideration arises in how the monthly FPU index is matched to each announcement. Because the index aggregates newspaper article counts over the full calendar month, its value for month t is not fully observable to investors at the time of an announcement released early in that month — it incorporates coverage published after the announcement date. The contemporaneous FPU value is therefore best interpreted not as a real-time information variable available to traders, but as an *ex-post classification* of the fiscal-uncertainty regime prevailing in the month surrounding the announcement. Under this interpretation it remains a valid conditioning variable for the question of whether announcement reactions differ across fiscal-uncertainty regimes, but it does not represent information investors could have acted upon in real time.

To address this directly, the analysis is conducted using two complementary measures of fiscal uncertainty. The first uses the contemporaneous monthly FPU value, interpreted as an ex-post regime classifier as described above. The second uses the prior month's FPU value (FPU_{t-1}), which is fully predetermined at the time of every announcement and therefore free of any within-month look-ahead, following Benchimol et al. (2023), who lag their uncertainty measure by one period for precisely this reason. The two specifications are reported alongside one another. The discussion leads with the contemporaneous specification, interpreted as an ex-post classifier of the fiscal-uncertainty regime prevailing around each announcement, and treats the lagged specification as a co-main robustness check that confirms the principal interactions are not an artefact of within-month look-ahead in the FPU timing. The mechanical endogeneity concern — that releases might themselves drive the index — is in any case minimal: the FPU index is built from fiscal-policy newspaper terms, none of which are directly triggered by the macroeconomic releases studied. The one partial exception, the Federal Budget Balance release (whose subject matter overlaps with FPU keywords such as 'federal budget' and 'budget deficit'), is negligible in practice, as it represents one of sixteen announcement types and any single-day influence is diluted across the full month of article counts.

Importantly, the fiscal policy sub-index does not explicitly use 'government shutdown' as a category-specific keyword. Shutdown-related articles may nevertheless enter the FPU index when they satisfy the baseline EPU criteria and contain fiscal-policy terms such as 'federal budget', 'budget battle', 'budget deficit', or 'government spending'. The FPU-based fiscal stress measure and the shutdown-based identification strategy employed in section 3.1.4 are therefore conceptually distinct but not mutually exclusive. The former captures the broader newspaper-based climate of fiscal policy uncertainty, while the latter identifies discrete episodes of acute fiscal dysfunction. Consistent findings across both measures therefore provide stronger cross-identification evidence than either measure alone.

TABLE 2 — ANNOUNCEMENT OBSERVATIONS BY FPU_STD BUCKET

Announcement	<0 SD		0–1 SD		≥1 SD	
	N	Mean ret (bps)	N	Mean ret (bps)	N	Mean ret (bps)
CPI	128	-3.36	30	3.1	24	0.24
Construction spending	128	-0.8	30	5.08	27	1.2
Consumer confidence	129	0.04	30	-1.33	27	-4.44
Consumer credit	127	0.99	30	-2.85	27	-1.55
Durable goods	128	0.44	30	-1.22	26	3.09
Existing home sales	129	1.04	30	0.38	27	1.73
Federal budget	129	-0.22	30	2.27	27	-1.01
GDP	44	1.13	7	-4.74	10	-5.29
ISM N-Mfg PMI	126	-0.01	30	0.3	25	-4.18
ISM PMI	128	-1.28	30	4.05	27	3.77
Industrial production	128	0.01	30	0.26	27	0.11
NFP	128	-0.2	30	-2.52	27	4.95
New home sales	129	0.13	30	2.55	24	-0.88
Retail sales	127	-1.93	30	-2.4	26	5.63
US International trade	129	0.66	30	-0.87	27	1.71
Unemployment rate	128	-0.2	30	-2.52	26	4.72
Total	1965	—	457	—	404	—

Note: FPU_std is the standardized fiscal policy uncertainty index. Buckets are defined as <0 SD (below-average FPU), 0–1 SD (moderate FPU), and ≥1 SD (elevated FPU). Mean ret is the mean signed [0,+5] return in basis points. Signed (rather than absolute) returns are shown here to display the directional pattern across FPU regimes; the pooled models in Section 3.3.1 use absolute returns, while the type-specific Model 4 uses signed returns as reported here

3.1.4. Government shutdown identification

The federal fiscal year of the United States begins on October 1. Agencies and programs that rely on annual appropriations must have received interim or full-year funding from Congress and the President by this date. When appropriations are not enacted in time, a lapse in appropriations occurs, during which the federal government initiates a shutdown of affected activities. Government shutdowns represent discrete, precisely dated episodes of acute fiscal stress, triggered by Congressional budget impasses rather than market developments. This makes them a complementary identification strategy to the continuous FPU measure: while the FPU captures the broad climate of fiscal policy uncertainty, shutdowns identify moments of outright fiscal dysfunction. If announcement reactions are amplified under both measures, the evidence for a fiscal uncertainty channel is strengthened. Shutdown episodes are identified using the History, Art and Archives of the House of Representatives. Episodes in which formal shutdown procedures were followed are classified as shutdowns. As several shutdowns commenced at midnight, the first calendar day on which federal activities were suspended is treated as the start date throughout, consistent with the Congressional Research Service.

The sample period from June 2010 to May 2026 encompasses seven government shutdown episodes in total, of which five contain at least one announcement observation during the active shutdown period, generating 61 active-shutdown observations in total across the sample. However, not all shutdowns coincide with scheduled macroeconomic announcement releases. This is partly because some shutdowns were too brief or fell between release dates, and partly because the statistical agencies responsible for producing the announcements studied here — such as the Bureau of Labor Statistics and the Census Bureau — are themselves subject to shutdown, meaning that releases scheduled during an active shutdown are often delayed or cancelled entirely rather than published on time. A recent example is the New Home Sales release, which is typically published within the final ten days of each month and reports activity from the preceding month. The October release, which would normally have been published in November, was delayed until January 13 due to the government shutdown of October–November 2025². By contrast, the CPI release for October 2025, scheduled for November 13, was cancelled entirely. To illustrate the broader pattern, across the three shutdown episodes of 2025–2026, 128 planned announcement releases were affected in total — including releases not covered by the present study. Of these, 95 were delayed, 20 were released on time, and 13 were cancelled entirely.³ This underscores why the post-shutdown window is an empirically relevant period for studying delayed information effects. The distribution of on time- and delayed releases across the sample of this study is provided in Table 4.

To capture uncertainty effects beyond the active shutdown window, a pre-shutdown window of 20 days and a post-shutdown window of 45 calendar days each are constructed around every

² See U.S. Census Bureau and U.S. Department of Housing and Urban Development, “Monthly New Residential Sales, October 2025,” Release Number CB26-09, released January 13, 2026. The release concerns October 2025 new residential sales, implying a delayed publication relative to the normal monthly release schedule.

³ See Paweł Skrzypczyński, “U.S. economic data schedule after the government shutdown,” updated May 7, 2026. The page compiles original, revised, and cancelled release dates for U.S. macroeconomic announcements affected by the 2025–2026 government shutdown episodes. The delayed, on-time, and cancelled counts reported here are the author’s own classification based on this schedule.

episode. The pre-window tests whether markets anticipate fiscal stress in the period leading up to a shutdown. The post-window captures delayed information effects: government shutdowns disrupt the statistical agencies responsible for producing labour market and activity data, causing delays in scheduled releases and potentially affecting data quality due to reduced staffing and incomplete data collection during the shutdown period. Announcements released in the 45 days following a shutdown may therefore belong to the shutdown period in economic terms, even though they are released after it officially ends — and markets may react more strongly to information that has been delayed or produced under disrupted conditions. Across the sample, 72 observations fall in the pre-window and 152 in the post-window. 20 observations fall within both a pre- and post-window, because consecutive shutdown episodes occurred close enough together that the post-window of one episode overlaps the pre-window of the next. These observations are retained in the main specifications and assigned to both windows simultaneously: economically, such a release genuinely belongs to both the aftermath of one shutdown and the run-up to the next, so the pre- and post-window indicators are not mutually exclusive and each overlapping observation contributes to both phase coefficients. As a robustness check, the analysis is re-estimated with these ten observations excluded, to confirm that the phase estimates are not driven by their double assignment. A descriptive overview of the shutdown episodes is provided in Table 3.

With only five distinct episodes containing announcement observations and 61 active-shutdown observations in total, the shutdown identification strategy still faces a power constraint, which shapes the modelling choices described in section 3.3.

TABLE 3 — GOVERNMENT SHUTDOWN EPISODES

ID	Start	End	Days	N observations (active)	N announcement types
18	01/10/2013	17/10/2013	17	3	3
19	20/01/2018	22/01/2018	3	0	0
20	09/02/2018	09/02/2018	1	0	0
21	22/12/2018	25/01/2019	35	9	9
22	01/10/2025	12/11/2025	43	10	7
23	31/01/2026	03/02/2026	4	1	1
24	14/02/2026	30/04/2026	76	38	16

Note: Only shutdowns from 2013 onwards fall within the sample period (2010–2026). N observations and N announcement types reflect active shutdown days only.

TABLE 4 — RELEASE TIMING BY SHUTDOWN PHASE

Phase	On time	Delayed	Total	Delayed (%)
Pre-window (20d)	58	14	72	19.4
Active shutdown	49	12	61	19.7
Post-window (45d)	103	49	152	32.2

Note: On time = announcements released on schedule. Delayed = announcements originally scheduled during the shutdown but released after its conclusion. Delayed (%) = share of delayed releases within each phase. Delayed releases within the pre-window are the result of two consecutive shutdowns.

Table 5 provides summary statistics for ES and ZN returns and announcement surprises across fiscal stress regimes and shutdown phases.

TABLE 5 — SUMMARY STATISTICS BY FISCAL UNCERTAINTY REGIME AND SHUTDOWN PHASE

	ES ret[0,+5] (bps)		ZN ret[0,+5] (bps)		Surprise		N
Regime / Phase	Mean	Std	Mean	Std	Mean	Std	
Fiscal policy uncertainty							
Non-stress	-0.41	23.18	-0.87	16.68	0.65	0.66	2,179
Fiscal stress	1.11	23.75	-0.08	14.76	0.74	0.83	676
Shutdown phases							
Pre-window	1.75	10.85	-1.46	11.89	0.63	0.68	72
Active shutdown	-1.85	14.16	-0.49	11.1	0.8	0.85	61
Post-window	1.43	24.24	-1.92	15.33	0.63	0.64	152

Note: ES and ZN returns are the [0,+5] minute returns (announcement-bar open-to-close) in basis points for S&P 500 and 10-year Treasury Note futures respectively. |Surprise| is the absolute standardized announcement surprise. Fiscal stress = FPU above 75th percentile.

3.2 Event Window Construction

The market reaction to each announcement is measured using a narrow intraday event window around the official release time. The baseline window is denoted $[0,+5]$, using the opening price and closing price of the bar at announcement. The resulting return isolates the price change of the start of the announcement until the close, five minutes after the announcement.

The use of intraday rather than daily returns follows the high-frequency announcement literature and is motivated by the rapid incorporation of public information in liquid futures markets. Under the efficient market hypothesis (Fama, 1970), prices adjust quickly to new information, implying that the surprise component of a scheduled macroeconomic announcement should be absorbed within a short interval around the release. Daily returns, often used in early literature, may incorporate other market-moving events occurring on the same trading day, making it harder to isolate the contribution of the announcement itself. A narrow intraday window therefore improves identification by reducing contamination from unrelated information.

The $[0,+5]$ window is conform with Andersen et al. (2003), who model exchange-rate responses using five-minute returns and show that conditional-mean adjustment occurs rapidly around scheduled macroeconomic releases. Using this strict post-announcements window, allows isolating the reaction, and excluding pre-announcement noise. To assess robustness to the event-window choice, four alternative windows are also computed: three progressively longer post-release windows, $[0,+10]$, $[0,+15]$, and $[0,+30]$, and a symmetric $[-5,+5]$ window spanning five minutes either side of the release. These allow verification that the results are not driven by the specific baseline window definition.

A portion of the announcements are released on the same date and at the same minute. Most importantly, the Employment Situation report is released as a single bundle containing both Nonfarm Payrolls and the Unemployment Rate, so these two series share an identical release timestamp; all 186 of their releases coincide. Similarly, the ISM Manufacturing PMI and Construction Spending are released at the same minute, with 168 coinciding releases. Because announcements sharing a release minute generate a single market return, treating each announcement series as a separate observation would assign that one return to multiple rows, mechanically duplicating the dependent variable, inflating the effective sample size, and producing standard errors that are too optimistic.

To avoid this, the unit of observation is defined as the *event-time* (a unique date-and-minute) rather than the announcement series. Each event-time carries a single return, and the surprises of all announcements released at that minute enter as separate regressors in the type-specific models (Section 3.3). Standard errors are clustered by event-time throughout, so that any residual correlation among announcements sharing a minute is accounted for.

3.3 Empirical Models

3.3.1 Fiscal Policy Uncertainty models

As a first step, a baseline linear regression model is estimated for both the equity market (ES) and the bond market (ZN). The dependent variable is the absolute return around the announcement window, while the key independent variable is the absolute standardized announcement surprise. The error term captures the component of returns not explained by the model. Because the objective of this initial specification is to test whether larger surprises generate larger market

reactions, rather than to estimate the directional effect of each announcement, both returns and surprises enter in absolute value.

$$(1) \quad |r_t| = \alpha + \beta_1 |S_t| + \epsilon_t$$

To examine whether market reactions differ during periods of high fiscal uncertainty, the standardized FPU is added as an independent variable, together with an interaction between the absolute announcement surprise and standardized FPU. This allows the interaction coefficient to capture how the sensitivity of market reactions to announcement surprises changes when fiscal policy uncertainty increases by one standard deviation.

$$(2) \quad |r_t| = \alpha + \beta_1 |S_t| + \beta_2 FPU_std_t + \beta_3 (|S_t| FPU_std_t) + \epsilon_t$$

Since macroeconomic announcements may differ systematically in their average market impact, failing to account for announcement type risks attributing baseline differences across releases to the interaction term. For example, Fleming and Remolona (1997) show that, in the U.S. Treasury market, employment releases generate substantially larger price-volatility effects than retail sales releases. To address this, announcement-type fixed effects, γ_j , are introduced in Model 3, absorbing type-specific differences in average market reactions before estimating the interaction between announcement surprises and FPU. This logic is analogous to Baker et al. (2021), who include firm fixed effects to absorb persistent cross-sectional differences in baseline volatility before estimating the effect of uncertainty exposure.

$$(3) \quad |r_t| = \alpha + \beta_1 |S_t| + \beta_2 FPU_std_t + \beta_3 (|S_t| FPU_std_t) + \gamma_j + \epsilon_t$$

While Model 3 controls for type-level intercept differences through announcement-type fixed effects γ_j , it imposes a single pooled surprise slope β_1 across all announcement types. This restriction is potentially problematic. If announcement types differ in their true surprise sensitivity — as the empirical literature strongly suggests, with labor market releases historically generating substantially larger market reactions than, for example, construction spending releases — the unmodelled slope heterogeneity is absorbed into the error term. Moreover, the composition of announcements may itself shift with fiscal stress. During a government shutdown, federally funded statistical agencies may suspend, delay, or cancel scheduled releases, so that a larger share of the announcements actually published during such periods originates from private institutions — for example, the ISM Manufacturing PMI, the ISM Non-Manufacturing PMI, and Consumer Confidence. If the remaining announcement types systematically carry a higher baseline surprise sensitivity, high-FPU periods would mechanically appear to exhibit stronger announcement reactions. The term FPU_std could then become correlated with the error term, raising a concern of omitted-variable bias that contaminates the estimated interaction coefficient β_3 . Model 4 addresses this by allowing each announcement type its own surprise slope β_j and its own FPU interaction δ_j , explicitly absorbing the slope heterogeneity and ensuring that the interaction estimates reflect genuine variation in fiscal uncertainty rather than announcement composition effects. This follows the approach of Flannery and Protopapadakis (2002), who assign each of their 17 macro announcement series a distinct return coefficient, recognizing that different releases carry fundamentally different information content for equity markets. Model 4 is therefore the

preferred specification, with model 2 and 3 serving as transparent intermediate steps that illustrate the sensitivity of the results to progressively relaxing the homogeneity restrictions imposed on the surprise-return relationship. This transition also changes the question being asked, not only the flexibility of the specification. Models 1 through 3 use absolute returns and absolute surprises, and therefore test whether larger surprises generate larger market reactions in magnitude, regardless of direction — that is, whether announcements move markets at all. Model 4 uses signed returns and signed surprises, and therefore tests the directional effect of each announcement type's surprise: whether a positive surprise raises or lowers the price, and by how much. The signed formulation is necessary because type-specific slopes capture directional effects, and pooling absolute magnitudes across types would obscure this heterogeneity — a positive NFP surprise and a positive CPI surprise have fundamentally different implications for equity and bond markets. These are therefore complementary but distinct questions: the pooled models establish that surprises move markets, while the type-specific signed model characterizes how. The Model 4 coefficients should accordingly be read as directional sensitivities in their own right, not as continuous extensions of the absolute-value estimates from Models 1 through 3

$$(4) \quad r_t = \alpha + \sum_j \gamma_j + \sum_j \beta_j S_{jt} + \beta_{FPU} FPU_{std} + \sum_j \delta_j (S_{jt} FPU_{std_t}) + \epsilon_t$$

Because Model 4 assigns each announcement type its own surprise slope, it also accommodates the event-time unit of observation defined in Section 3.2. When two or more announcements share a release minute, the single market return for that event-time is regressed jointly on the separate surprises, each entering with its own slope and its own FPU interaction. For the Employment Situation release, which bundles Nonfarm Payrolls and the Unemployment Rate at the same timestamp, this takes the form:

$$(4a) \quad r_t = \alpha + \gamma_j + \beta_{NFP} S_{NFP,t} + \beta_{UR} S_{UR,t} + \beta_{FPU} FPU_{std} + \delta_{NFP} (S_{NFP,t} FPU_{std_t}) + \delta_{UR} (S_{UR,t} FPU_{std_t}) + \epsilon_t$$

To test the robustness of model 4, a binary fiscal stress regime is constructed by applying a threshold to the raw FPU series. Observations above the threshold are classified as high fiscal stress periods (dummy = 1) and observations below as normal periods (dummy = 0). The dummy replaces FPU_std in an otherwise identical type-specific interaction model, providing a regime-based interpretation that is less sensitive to the linearity assumption embedded in the continuous specification. To ensure the results are not driven by the choice of threshold, the analysis is repeated at the 70th, 75th, 80th, and 90th percentiles of the FPU distribution, with the 75th percentile serving as the primary robustness benchmark.

All continuous FPU and dummy robustness specifications are estimated in parallel for 10-year Treasury Note futures (ZN), using the same [0,+5] minute return window (announcement-bar open-to-close) applied to ZN closing prices. The ZN models replicate the ES structure exactly, with the ZN return replacing the ES return as the dependent variable while all regressors remain unchanged. Estimating identical specifications across both markets serves two purposes. First, it tests whether the FPU effects documented for equities extend to fixed income, providing a cross-market validation of the main findings. Second, and more importantly, it allows identification of announcement types where the ES and ZN interaction coefficients move in opposite directions under fiscal stress. Such a pattern — where high FPU simultaneously dampens equity sensitivity and amplifies Treasury sensitivity to the same surprise — is consistent with flight-to-safety dynamics, and represents a market-level mechanism that a pure noise or attention story cannot

easily explain. Because ZN is a futures price, a positive ZN return corresponds to a fall in Treasury yields. A configuration in which a surprise lowers the equity return while raising the ZN return therefore reflects a simultaneous decline in equities and yields, consistent with flight-to-safety dynamics.

3.3.2 Shutdown models

The shutdown models follow the same structure as the FPU models, progressing from a pooled specification to one that allows type-specific interaction coefficients.

Government shutdowns have fixed start and end dates, but the period of elevated fiscal uncertainty likely extends beyond the active episode itself. Shutdowns can be anticipated by markets: as negotiations approach the fiscal deadline of 1 October and show signs of deterioration, the probability of a shutdown becomes increasingly visible to investors. The exact timing and duration, however, remain unknown. This combination of anticipatable onset and uncertain resolution means that fiscal uncertainty may begin rising before the shutdown officially starts. Bacon & Woodard (2015) confirms this, documenting significant abnormal trading activity up to 30 days before a shutdown announcement. Following Veronesi (1999), who shows that investors' price reactions to news are amplified when they face uncertainty about the prevailing economic regime, this anticipatory uncertainty provides a theoretical basis for expecting altered announcement sensitivity in the days preceding a shutdown. A pre-window of 20 calendar days is therefore defined to capture this period.

It is equally plausible that uncertainty does not resolve simultaneously with the end of the shutdown. Fiscal stress episodes often leave unresolved questions about budget trajectories, debt ceiling negotiations, and future political stability. Moreover, government statistical agencies suspend data releases during shutdowns, creating a backlog of delayed announcements. Once the shutdown ends, markets face an unusually information-dense environment in which the informational value of each release may differ from normal periods. A post-window of 45 calendar days is defined to capture this elevated post-shutdown period.

Three phases are thus identified: pre-shutdown, active shutdown, and post-shutdown. A pooled linear regression is first run on each phase separately to test whether announcement reactions differ from non-episode periods at all. The three phases are then combined into a single episode dummy to assess whether the shutdown episode as a whole generates an aggregate effect on announcement sensitivity, before a joint phase model decomposes this into simultaneous phase-specific coefficients, allowing direct comparison of the three phases while controlling for each other.

Following the same logic as the transition from model 3 to model 4, each phase-specific model is extended to allow type-specific interaction coefficients. This yields the post-window model and the pre-window model presented below — the type-specific extension is warranted for the same reasons as in the FPU context: pooling across announcement types risks masking heterogeneous phase effects and introduces composition bias if certain types are systematically over- or under-represented within a given phase.

$$\begin{aligned} \text{Post model:} \quad & r_t = \alpha + \sum_j \gamma_j + \sum_j \beta_j S_{jt} + \beta_{post} post_t + \sum_j \delta_j (S_{jt} post_t) + \epsilon_t \\ \text{Pre model:} \quad & r_t = \alpha + \sum_j \gamma_j + \sum_j \beta_j S_{jt} + \beta_{pre} pre_t + \sum_j \delta_j (S_{jt} pre_t) + \epsilon_t \end{aligned}$$

Finally, the post-window contains two distinct types of releases: announcements released on schedule, and announcements that were originally planned during the shutdown but delayed until after its conclusion. The post-window model captures both jointly, but the mechanisms underlying each may differ — delayed releases carry information that markets were specifically denied during the shutdown, and their eventual release may trigger stronger reactions as investors update on a larger information backlog. A separate model isolating confirmed delayed releases is therefore estimated to test this information substitution mechanism directly. All shutdown models are estimated for both the S&P 500 futures (ES) and 10-year Treasury Note futures (ZN), following the same parallel structure as the FPU specifications in section 3.3.1.

3.3.3 Additional analyses

To examine whether the direction of an announcement surprise matters beyond its magnitude, an asymmetry analysis is conducted. A positive surprise indicator is interacted with the fiscal stress dummy and the announcement surprise, yielding a three-way interaction term that captures whether positive surprises are amplified or dampened differently from negative surprises during periods of high fiscal uncertainty. This is estimated first in a pooled specification (MA1) and then with announcement-type fixed effects (MA2), following the same progression as the main FPU models.

To assess the robustness of the main findings at the individual announcement level, separate regressions are estimated for each of the 16 announcement types, regressing returns on the surprise, the fiscal stress indicator, and their interaction.

All models are estimated using heteroskedasticity-consistent standard errors (HC3), which provide valid inference without imposing assumptions on the error variance structure. Because the unit of observation is the event-time (Section 3.2), and announcements released at the same minute share a single market return, the type-specific models are estimated with standard errors clustered by event-time as the primary specification; this accounts for any residual correlation among announcements that share a release window. HC3 standard errors are additionally reported for these models, and where the two differ materially both are shown. The clustered and HC3 results are generally close, with clustering yielding slightly wider intervals, as expected once the shared-return structure is accounted for.

The shutdown specifications warrant particular caution in interpretation. While the total number of announcement observations entering these models is substantial, the identifying variation comes from a small number of distinct shutdown episodes — only five contain announcement observations during the active-shutdown window, with correspondingly few episodes in the pre- and post-windows. With so few independent episodes, the effective degrees of freedom are limited, and the standard errors, however computed, may understate the true sampling uncertainty. A wild-cluster bootstrap was considered as an alternative means of inference, but with only five clusters its behaviour is itself unreliable, as the procedure requires a sufficient number of clusters to approximate the sampling distribution well. The shutdown estimates — and especially the type-specific coefficients for thinly populated announcement types such as GDP — should therefore be read as suggestive rather than definitive, and are interpreted accordingly in the results.

4. Empirical results

This section presents the empirical results. Fiscal stress is identified in two complementary ways: through a continuous and threshold-based measure of fiscal policy uncertainty built from the FPU index, and through discrete government shutdown episodes. For each, the analysis begins with pooled specifications that estimate a single average effect across all announcements, then moves to type-specific models that give each announcement its own surprise sensitivity and fiscal-stress interaction, estimated for both equity (ES) and Treasury (ZN) futures. The central finding is that fiscal uncertainty does not act uniformly on announcement reactions. Rather than scaling all reactions up or down, it reshapes sensitivity selectively. Fiscal uncertainty appears to activate equity reactions to several labour-market and real-activity releases that are not significant under baseline conditions, in the sense that their baseline surprise slopes are insignificant while their FPU interaction terms are statistically significant. It amplifies the reaction to a few others and dampens the bond market's response to real-activity surprises. For a subset of announcements — most clearly Retail Sales — equity and Treasury markets move in opposite directions under fiscal stress, a pattern consistent with flight-to-safety. The government-shutdown analysis offers a complementary, event-based perspective, though its estimates rest on few episodes and are interpreted as suggestive.

4.1 Fiscal policy uncertainty and announcement reactions

Table 6 presents Models 1–3. Model 1 regresses the absolute return on the pooled absolute surprise across all 2,855 announcements. Larger surprises generate significantly larger market reactions ($\beta = 1.93$, SE 0.55, $p < 0.01$). The null of no reaction ($\beta_1 = 0$) is therefore rejected, confirming Hypothesis 1. The low adjusted R^2 (0.005) is expected: a single common slope across all announcement types cannot capture the cross-type heterogeneity that motivates Model 4, and the pooled specifications are reported to test coefficient significance rather than fit. This concerns only the magnitude of the reaction, not its direction. The finding is consistent with the macro-announcement literature and establishes the foundation for the analysis.

TABLE 6 — BASELINE AND POOLED FPU MODELS (ES)

	Model 1	Model 2	Model 3
Surprise	1.93*** (0.55)	1.93*** (0.57)	1.86*** (0.55)
FPU		1.18*** (0.44)	1.06*** (0.38)
Surprise $\times$ FPU		-0.52 (0.44)	-0.31 (0.40)
Dependent variable	return	return	return
Announcement FE	No	No	Yes
SE type	HC3	Event-time	Event-time
N	2855	2826	2826
Adj. R^2	0.005	0.006	0.160

*Note: Models 1–3 use absolute returns and absolute standardized surprises. Coefficient (bps) with standard error in parentheses. Model 1 uses HC3 standard errors; Models 2–3 cluster by event-time. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.*

Model 2 adds FPU and its interaction with the surprise. The surprise effect is essentially unchanged ($\beta_1 = +1.93$ bps, $p < 0.01$). Fiscal uncertainty carries a significant positive level effect of 1.18 bps ($\beta_2 = +1.18$, $p < 0.01$), meaning reaction magnitudes are larger when fiscal uncertainty is high. The interaction, however, is insignificant ($\beta_3 = -0.52$, $p > 0.10$), and remains so once announcement-type fixed effects are added in Model 3 ($\beta_3 = -0.31$, $p > 0.10$), where the FPU level effect stays significant at 1.06 bps ($p < 0.01$) and the adjusted R^2 rises to 0.160. The null that $\beta_3 = 0$ therefore cannot be rejected. At the pooled level, therefore, Hypothesis 2 is not supported: fiscal uncertainty does not significantly amplify the average surprise-return sensitivity when a single common slope is imposed across announcement types. This null is itself informative, as it points to the restriction rather than the absence of an effect — a single pooled slope averages amplification in some announcements against dampening in others. Whether such offsetting heterogeneity is in fact present is the question Model 4 addresses.

Model 4 (using contemporaneous FPU) assigns each announcement its own slope and uses signed returns and surprises, capturing directional effects. Inflation and forward-looking survey indicators generate the strongest equity market reactions. A one-standard-deviation (SD) increase in the CPI surprise moves the S&P down 23.8 basis points on average ($p < 0.01$). This operates through a monetary-policy anticipation channel: a positive inflation surprise raises expected policy rates and hence the discount rate applied to equities, lowering valuations. Bond markets, in addition, react to real-activity and labour-market surprises where the S&P does not respond significantly. While NFP and Unemployment Rate surprises produce insignificant reactions for ES in the baseline specification, both generate significant reactions for ZN. A one-SD increase in the NFP surprise produces a 15.8 bps decline in ZN on average ($p < 0.01$), the largest ZN reaction among the announcements studied. This is consistent with the Phillips-curve interpretation in Balduzzi et al. (2001): stronger-than-expected employment growth signals stronger real activity, which can raise expected inflation and future short rates, lifting yields and lowering Treasury futures prices. The reaction to Unemployment Rate surprises is considerably smaller, also in line with Balduzzi et al., who find that Nonfarm Payrolls carry substantially more bond-market-relevant information than the unemployment-rate component of the employment report. Notably, Balduzzi et al. point out that because Nonfarm Payrolls and the Unemployment Rate are released together at 8:30, their influences cannot be separated without the individual surprise components; the joint specification adopted here, which enters both surprises as separate regressors on the shared employment-report return, recovers precisely this distinction and confirms that the bond-market reaction is driven predominantly by the payrolls component. Among the announcement surprises that produce significant reactions in both markets, the signs of the reaction coefficients are opposite for ES and ZN in eight of nine cases. CPI is the sole exception: a positive inflation surprise is bad for equities, through the higher discount rate described above, and bad for bonds, as higher expected inflation raises yields and lowers prices, producing a reaction of the same sign in both markets.

TABLE 7 — TYPE-SPECIFIC FPU INTERACTIONS (MODEL 4)

	S&P 500 (ES)		10-Yr T-Note (ZN)	
	β_j	$\delta_j (\times \text{FPU})$	β_j	$\delta_j (\times \text{FPU})$
<i>Real activity</i>				
NFP	3.97 (2.67)	9.71** (4.38)	-15.80*** (3.23)	3.01 (3.53)
Unemployment rate	2.30 (2.45)	-9.04*** (3.24)	4.41* (2.26)	0.66 (2.91)
Retail sales	3.78** (1.56)	-3.92** (1.73)	-3.35** (1.51)	4.80** (1.89)
Industrial production	0.59 (0.48)	-1.11 (0.87)	-1.12*** (0.26)	0.60* (0.34)
Consumer credit	0.47 (0.49)	-1.20*** (0.42)	-0.26 (0.17)	-0.12 (0.13)
<i>Consumption</i>				
New home sales	3.41** (1.59)	0.17 (1.17)	-1.04* (0.58)	0.47 (0.41)
Existing home sales	2.44** (1.02)	3.05** (1.42)	-1.36*** (0.41)	-0.02 (0.40)
<i>Investment</i>				
Durable goods	2.09* (1.19)	-0.12 (0.63)	-2.03** (0.89)	-0.20 (0.36)
Construction spending	-1.53 (1.33)	-4.77*** (1.79)	0.21 (0.61)	1.59* (0.84)
<i>Output</i>				
GDP	5.86*** (1.59)	0.02 (1.63)	-3.78*** (1.35)	2.47* (1.40)
<i>Government</i>				
Federal budget	-0.93 (0.99)	1.35 (1.87)	0.14 (0.29)	-0.48 (0.31)
<i>Net exports</i>				
US International trade	-0.21 (1.13)	-0.32 (0.49)	-0.27 (0.89)	-0.08 (0.41)
<i>Prices</i>				
CPI	-23.80*** (5.29)	3.54 (2.97)	-12.49*** (2.25)	3.41* (1.94)
<i>Forward-looking</i>				
Consumer confidence	3.95*** (1.06)	1.53 (1.06)	-2.52*** (0.56)	0.29 (0.40)
ISM PMI	11.77*** (1.55)	6.16** (2.97)	-7.93*** (0.79)	0.35 (0.68)
ISM N-Mfg PMI	3.11*** (1.15)	2.08 (1.51)	-6.17*** (0.81)	0.24 (0.84)
Announcement FE	Yes		Yes	
SE type	Event-time		Event-time	
N	2367		2365	
Adj. R ²	0.140		0.216	

Note: β_j is the announcement-specific surprise slope; δ_j the interaction with standardized FPU (effect of a one-SD increase in fiscal policy uncertainty). Coefficient (bps), standard error in parentheses. Event-time clustered SE. A positive ZN return corresponds to a fall in Treasury yields. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Fiscal uncertainty does not amplify market reactions to macroeconomic surprises uniformly; it reshapes sensitivity heterogeneously, strengthening some reactions and dampening others. For the equity market, two distinct patterns appear. First, fiscal uncertainty significantly changes ES sensitivity to several real-activity releases whose baseline coefficients are insignificant, suggesting these announcements become more relevant for equity pricing under fiscal stress. The reaction of ES to NFP surprises is significant and positive under fiscal uncertainty ($\delta_{NFP} = +9.71$, $p < 0.05$) but only insignificantly positive under normal conditions. The same holds for the Unemployment Rate ($\delta_{UR} = -9.04$, $p < 0.01$) and Consumer Credit ($\delta_{CC} = -1.20$, $p < 0.01$), both with insignificant baseline ES slopes but significant interactions. For the Unemployment Rate, the negative interaction is economically intuitive because a positive unemployment surprise represents weaker-than-expected labour-market conditions, while a positive NFP surprise indicates stronger-than-expected labour-market conditions. Second, for releases that *already* generate a significant equity reaction, fiscal uncertainty amplifies the response. ES reacts significantly and positively to Existing Home Sales surprises in both normal and high-uncertainty periods ($\beta_{EHS} = +2.44$, $p < 0.05$; $\delta_{EHS} = +3.05$, $p < 0.05$), and the same amplification appears for the ISM PMI ($\beta_{PMI} = +11.77$, $p < 0.01$; $\delta_{PMI} = +6.16$, $p < 0.05$) — in both cases the interaction shares the sign of the baseline, strengthening the existing reaction.

Whereas fiscal uncertainty broadens the set of announcements the equity market responds to, the opposite holds for the bond market. While the bond market reacts significantly to four of the five real-activity surprises in the baseline, only two remain significant under fiscal uncertainty. Among real-activity surprises with a significant baseline bond reaction, fiscal uncertainty dampens the response. Industrial Production and GDP both carry interaction coefficients of opposite sign to their baseline, pulling the reaction toward zero. Industrial Production's negative baseline ZN slope ($\beta_{IP} = -1.12$, $p < 0.01$) is offset by a positive interaction ($\delta_{IP} = +0.60$, $p < 0.10$), and GDP's negative baseline ($\beta_{gdp} = -3.78$, $p < 0.01$) by a positive interaction ($\delta_{gdp} = +2.47$, $p < 0.10$). Construction Spending has no significant baseline bond reaction, and Retail Sales follows the distinct flight-to-safety pattern discussed below.

Retail Sales exhibits the clearest cross-market pattern. Under fiscal uncertainty, a positive Retail Sales surprise generates a negative reaction in the equity market ($\delta_{RS} = -3.92$, $p < 0.05$) and a positive reaction in the bond market ($\delta_{RS} = +4.80$, $p < 0.05$) — equities sell off while Treasury prices rise. Construction Spending displays the same configuration, though the bond-side evidence is weaker: a negative equity interaction ($\delta_{CS} = -4.77$, $p < 0.01$) alongside a positive bond interaction that is significant only at the 10% level ($\delta_{CS} = +1.59$, $p < 0.10$). Both patterns are consistent with a flight-to-safety mechanism. Baele et al. (2020) show that flight-to-safety episodes involve negative equity returns, positive government-bond returns, and flows out of equity funds into government bond and money market funds, and they associate these episodes with heightened market stress and uncertainty. In the present setting, fiscal stress may serve as the uncertainty shock generating a pricing pattern consistent with risk-off reallocation from equities toward Treasury exposure. This cross-market configuration confirms Hypothesis 4: for a subset of announcement types — most clearly Retail Sales, and more tentatively Construction Spending — equity and Treasury markets respond in opposite directions to the same surprise under fiscal stress, consistent with flight-to-safety dynamics. Because this pattern operates at the market level rather than within a single series, it is difficult to reconcile with a pure noise- or attention-based account, which would not predict a coordinated, opposite-signed reallocation across the two markets.

The interactions reported above are estimated using the contemporaneous monthly FPU. Because this value incorporates news published after early-month announcements, the

contemporaneous specification is an ex-post regime classification; the prior-month (lagged) specification is therefore treated as co-main. The principal equity interactions persist under this real-time-valid measure: NFP ($\delta_{NFP} = +8.63$, $p < 0.05$), the Unemployment Rate ($\delta_{UR} = -5.41$, $p < 0.05$), Consumer Credit ($\delta_{CC} = -0.93$, $p < 0.10$), Retail Sales ($\delta_{RS} = -4.98$, $p < 0.01$) and ISM PMI ($\delta_{PMI} = +5.29$, $p < 0.05$) all remain significant with unchanged signs. Existing Home Sales and Construction Spending lose significance under the lagged specification ($\delta_{EHS} = +1.84$ and $\delta_{CS} = -2.14$, both ns) and are interpreted with more caution.

Taken together, these results confirm Hypothesis 2a: the effect of fiscal uncertainty on announcement reactions is not uniform but differs markedly across announcement types — activating reactions to some, amplifying others, dampening others still, and producing a cross-market flight-to-safety pattern for a subset.

Recasting fiscal uncertainty as a discrete high-stress regime, by creating a dummy equal to one for months above the 75th percentile of the monthly FPU distribution, broadly corroborates the continuous findings. The core equity effects persist with unchanged signs: the Unemployment Rate ($\delta_{UR} = -9.85$, $p < 0.05$), Consumer Credit ($\delta_{CC} = -2.93$, $p < 0.05$), Construction Spending ($\delta_{CS} = -7.56$, $p < 0.05$) and ISM PMI ($\delta_{PMI} = +12.80$, $p < 0.01$) all remain significant. A few weaker effects, notably Retail Sales, lose significance under the regime definition, while others, Industrial Production ($\delta_{IP} = -1.87$, $p < 0.10$) and CPI ($\delta = +13.78$, $p < 0.10$), emerge — reflecting the different ways the two specifications weight the fiscal-uncertainty distribution. The NFP interaction in fact grows in magnitude ($\delta_{NFP} = +13.84$) but turns insignificant, consistent with the smaller number of observations in the stressed regime. This regime-based framing motivates a sharper, event-based identification of acute fiscal stress: government shutdowns.

4.2 Government shutdowns

Government shutdowns provide a discrete, event-based counterpart to the continuous FPU measure, marking episodes of acute fiscal stress. This section examines whether announcement sensitivity is altered in three windows defined around a shutdown — the pre-shutdown, active shutdown, and the post-shutdown windows — and whether releases delayed by a shutdown trigger different reactions upon publication. Because the sample contains few shutdown episodes, these estimates are interpreted as suggestive rather than definitive.

Table 8 reports five pooled equity-market shutdown models. Each of the first three isolates one window, while the latter two combine them. The first three present the estimation results for the active-, pre-, and post-shutdown windows. The fourth model, episode, combines the three phases and treats the combined period as a single window. Finally, the joint model estimates all three phase interactions simultaneously. In contrast to what was expected, the equity market reaction to announcement surprises is dampened during the 20-day window preceding the government shutdown. ($\delta_{pre} = -2.95$, $p < 0.05$). This effect is also present in the joint model, however the evidence is weaker ($\delta_{pre} = -2.88$, $p < 0.10$). This rejects the null of no pre-window effect ($\delta_{pre} = 0$) but contradicts H3b, which predicted amplified rather than dampened sensitivity ahead of a shutdown. The active- and post-shutdown windows show no significant pooled effect ($\delta_{active} = -1.06$ ns, $\delta_{post} = +0.47$, $p > 0.10$). For the active window this is consistent with H3a, which anticipated a weak effect given the limited number of active-shutdown observations; the post-window result provides no support for H3c. As with the pooled FPU models, however, these specifications impose a single common slope across all announcement types, which can mask offsetting heterogeneity: an insignificant pooled effect does not preclude significant type-specific

reactions within a window. The type-specific specification in Table 9 relaxes this restriction by estimating a separate phase interaction for each announcement type.

TABLE 8 — POOLED SHUTDOWN MODELS (ES)

	Active	Pre	Post	Episode	Joint
Surprise × Active	-1.06 (2.06)				-1.89 (3.71)
Surprise × Pre		-2.95** (1.24)			-2.88* (1.68)
Surprise × Post			0.47 (2.96)		1.85 (3.87)
Surprise × Episode				0.63 (1.98)	
SE type	Event-time	Event-time	Event-time	Event-time	Event-time
N	2856	2856	2856	2856	2856
Adj. R ²	0.005	0.006	0.004	0.005	0.006

*Note: Pooled shutdown models on absolute returns and absolute surprises. Columns Active, Pre, and Post report separate single-phase regressions (M6, M8, M9); Episode (M10) uses a combined episode dummy; Joint (M11) estimates all three phase interactions simultaneously. Coefficient (bps), SE in parentheses, event-time clustered. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.*

Table 9 reports the type-specific decomposition of the pre- and post-shutdown window effects. The active window is not estimated at the type level: with roughly sixty observations distributed very unevenly across announcements (as few as one for GDP), type-specific coefficients within the active window cannot be reliably identified, and the active shutdown is therefore assessed only at the pooled level in Table 8. The pre- and post-window estimates likewise rest on few observations per announcement — four to six in the pre-window (only one for GDP) and eight to twelve in the post-window — and several pre-window coefficients are correspondingly large and imprecise. The Unemployment Rate pre-window interaction of -69.64 bps, for instance, is estimated from four observations and far exceeds any baseline reaction in Table 7. The type-specific shutdown results are therefore exploratory and interpreted with caution; the pooled estimates in Table 8 remain the more reliable shutdown evidence. These estimates should not be interpreted as stable announcement-specific shutdown effects; they are reported only to document possible heterogeneity behind the pooled results.

With this caveat, the post-window estimates, which rest on more observations, are somewhat more trustworthy than the pre-window. The clearest type-specific effect is the post-window equity reaction to NFP surprises ($\delta +16.06$, $p < 0.10$), estimated from twelve observations, suggesting that payroll surprises released after a shutdown may elicit a stronger equity reaction — though the marginal significance and small sample warrant caution. No coherent type-specific pattern emerges beyond this, and the broad dispersion of coefficients is consistent with the limited number of shutdown episodes underlying the analysis.

TABLE 9 — TYPE-SPECIFIC SHUTDOWN INTERACTION COEFFICIENTS

	Pre-window (δ_i)		Post-window (δ_j)	
	ES	ZN	ES	ZN
Real activity				
NFP	-14.00 (12.42)	-29.16 (18.99)	16.06* (9.01)	-8.55 (9.64)
Unemployment rate	-69.64*** (15.86)	35.86* (20.76)	8.62 (10.49)	19.26 (11.95)
Retail sales	-6.13 (4.38)	-9.70* (4.96)	6.04 (4.60)	-3.89 (2.76)
Industrial production	-0.98 (3.77)	-1.04 (2.29)	0.53 (1.32)	-0.74 (1.42)
Consumer credit	-3.98 (3.90)	-1.01 (0.93)	-2.61 (2.34)	1.32 (0.98)
Consumption				
New home sales	-3.43 (2.30)	-0.40 (0.76)	0.22 (2.29)	0.27 (0.92)
Existing home sales	-3.22 (2.64)	-1.90 (2.56)	0.34 (1.98)	1.07 (0.82)
Investment				
Durable goods	-7.77*** (2.86)	-8.76*** (2.86)	1.91 (4.32)	5.05 (3.54)
Construction spending	0.44 (4.14)	-3.52* (2.08)	3.02 (6.81)	-2.59* (1.54)
Output				
GDP	-9.95 (7.22)	-26.11*** (6.22)	-2.85 (3.61)	-0.34 (3.06)
Government				
Federal budget	3.89 (3.15)	0.34 (1.86)	3.19 (2.19)	-2.26 (1.61)
Net exports				
US International trade	0.32 (0.87)	-1.37** (0.64)	-1.52 (1.15)	0.27 (0.66)
Prices				
CPI	10.61 (10.22)	23.40** (10.43)	-35.28 (23.89)	0.32 (7.41)
Forward-looking				
Consumer confidence	-5.53* (2.85)	0.63 (2.07)	7.06 (5.94)	0.81 (1.40)
ISM PMI	-5.49** (2.27)	1.38 (1.40)	5.94 (6.22)	-1.18 (1.52)
ISM N-Mfg PMI	2.18 (4.10)	4.92*** (1.41)	2.57 (2.84)	3.86** (1.77)
N	2394	2392	2394	2392
Adj. R ²	0.112	0.209	0.120	0.205

Note: Type-specific interaction coefficients δ_i between the announcement surprise and the pre-window (M13_pre) or post-window (M13) indicator, estimated jointly with announcement fixed effects. Coefficient (bps), SE in parentheses, event-time clustered. A positive ZN return corresponds to a fall in Treasury yields. Shutdown estimates rest on few episodes and should be read as suggestive. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

4.3 Delayed releases

Table 10 reports the market's reaction to delayed releases, based on a pooled model. Delayed releases show no significantly different baseline reaction (level effect -1.43 , ns), but the equity market's reaction *to the surprise* is significantly dampened. The announcement surprise generates a significant positive reaction on ES over the full sample ($\beta = +2.07$, $p < 0.01$), while the interaction with delayed releases is significantly negative ($\delta = -2.37$, $p < 0.01$). This pulls the average reaction to a delayed surprise toward zero, corresponding to an overall dampened reaction. The same model estimated on ZN shows no significant delayed-release interaction (-0.52 , $p > 0.10$), indicating the dampening is equity-specific. The significant interaction rejects the null of no delayed-release effect on the equity surprise reaction ($\delta = 0$); the effect is negative, however, contradicting H3d, which predicted amplification through an information-backlog mechanism. Rather than reacting more strongly, the equity market reacts less to the surprise content of delayed releases. Several mechanisms could account for this dampening: by the time a delayed release is published, much of its information content may already have reached the market through other channels, leaving a smaller surprise to be priced; the eventual release may be partly anticipated, having been expected since the shutdown's resolution; or data compiled under disrupted conditions may be regarded as lower quality and weighted less heavily.

TABLE 10 — POOLED DELAYED-RELEASE MODEL (ES)

	Coefficient
Surprise	2.07*** (0.57)
Delayed release	-1.43 (1.60)
Surprise $\times$ Delayed	-2.37*** (0.71)
Announcement FE	Yes
SE type	Event-time
N	2856
Adj. R ²	0.158

*Note: Pooled test of the information-substitution hypothesis on absolute returns and absolute surprises with announcement fixed effects. "Delayed release" equals one for announcements originally scheduled during a shutdown but published after it ended. The |Surprise| $\times$ Delayed interaction tests whether the surprise reaction differs for delayed releases. Coefficient (bps), event-time clustered SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.*

4.4 Robustness

The principal findings are robust to a range of alternative specifications. Replacing event-time clustering with HC3 standard errors leaves the type-specific FPU interactions materially unchanged (Table A1). Standardizing surprises on forecast errors winsorized at the 1st and 99th percentiles, which limits the influence of extreme surprises, preserves the same pattern of

significant effects (Table A3). The results are not an artefact of the chosen event window: re-estimating the main model over $[0,+10]$, $[0,+15]$, $[0,+30]$ and the symmetric $[-5,+5]$ window yields consistent conclusions, with the $[0,+5]$ window producing the largest number of significant interactions (Table A4). Finally, computing the surprise standard deviation over the full sample including the COVID-19 period leaves the signs and statistical significance of all principal interactions unchanged, though the size of the coefficients changes because the larger standard deviation rescales the surprise variable.

5. Conclusion and future research

This thesis examined whether fiscal uncertainty conditions the intraday reaction of S&P 500 futures to macroeconomic announcement surprises — a question that, despite an extensive literature on announcement effects and on uncertainty more broadly, had not previously been studied. Using two complementary identification strategies, a continuous measure of fiscal policy uncertainty built from the fiscal-policy component of the Baker et al. (2016) index and discrete U.S. government shutdown episodes, the analysis produced a consistent central finding: fiscal uncertainty does not scale announcement reactions uniformly, but reshapes them selectively.

The baseline results confirm that larger surprises generate larger market reactions (H1), establishing the foundation for the conditional analysis. At the pooled level, fiscal uncertainty does not significantly amplify the average surprise-return sensitivity (H2 is not supported), but this null masks substantial heterogeneity: once each announcement type is given its own sensitivity, the effect of fiscal uncertainty differs markedly across release types (H2a confirmed). Fiscal uncertainty activates equity reactions to several labour-market and real-activity releases that are insignificant under normal conditions, amplifies the reaction to a few others, and dampens the bond market's response to real-activity surprises. For a subset of releases — most clearly Retail Sales — equity and Treasury markets move in opposite directions under fiscal stress, a pattern consistent with flight-to-safety (H4 confirmed). The government shutdown analysis offers a complementary, event-based perspective. Announcement sensitivity is significantly dampened in the pre-shutdown window, contrary to the amplification that had been anticipated (H3b rejected); the active and post-shutdown windows show no significant pooled effect (H3a uninformative given limited power; H3c unsupported); and the reaction to delayed releases is dampened rather than amplified (H3d rejected). Taken together, the two strategies point to the same conclusion: fiscal stress changes how markets process macroeconomic information, but it does so selectively and through channels that differ from the amplification documented for economic uncertainty.

These findings contribute to the literature in three ways. They extend the uncertainty-announcement reaction literature to a fiscal uncertainty regime that had not previously been examined; they move the study of government shutdowns beyond price-level and volatility effects toward the information-processing mechanism; and they suggest that fiscal dysfunction creates a distinct conditioning environment, rather than operating through the policy-anticipation or informational-value channels identified in prior work.

Several limitations qualify these results and point toward future research. The most important constraint concerns the government shutdown analysis. The identifying variation comes from only a small number of distinct episodes, with correspondingly few announcement observations during active-shutdown, pre-, and post-windows. The effective degrees of freedom are therefore limited, the type-specific shutdown estimates are imprecise, and the shutdown results are interpreted

throughout as suggestive rather than definitive. Future research could substantially strengthen this identification by extending the sample backward in time. The Money Market Services (MMS) survey data, which provide consensus forecasts reaching back well before the start of the present sample, would allow the inclusion of additional government shutdown episodes from earlier decades, increasing the number of shutdown observations and improving the precision of the event-based estimates.

A second limitation is that the present study conditions announcement reactions on fiscal uncertainty alone, without controlling for the broader state of the economy. Yet the literature establishes that announcement reactions are themselves state-dependent: the same surprise can move markets in opposite directions depending on whether the economy is in expansion or contraction (McQueen & Roley, 1993; Boyd et al., 2005). To the extent that fiscal stress episodes coincide with particular phases of the business cycle, part of what is attributed here to fiscal uncertainty may reflect the underlying economic state. Future research could incorporate the state of the economy — for example through recession indicators or business-cycle measures — as an additional conditioning variable, yielding more accurate estimates of the fiscal uncertainty effect and disentangling it from the well-documented influence of the business cycle.

Finally, the analysis focuses on equity and Treasury futures and on a single, media-based measure of fiscal uncertainty. Future work could extend the approach to other asset classes, or complement the newspaper-based fiscal policy index with alternative measures of fiscal stress, to test the robustness of the selective-reshaping mechanism documented here. Notwithstanding these limitations, the central result is robust across both identification strategies and a range of alternative specifications: fiscal uncertainty is a distinct and economically meaningful conditioning regime for the pricing of macroeconomic information.

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7. Appendix

TABLE A1 — MODEL 4 WITH HC3 STANDARD ERRORS (ES)

	δ_j (×FPU)
<i>Real activity</i>	
NFP	9.71** (4.55)
Unemployment rate	-9.04*** (3.39)
Retail sales	-3.92** (1.86)
Industrial production	-1.11 (0.92)
Consumer credit	-1.20*** (0.46)
<i>Consumption</i>	
New home sales	0.17 (1.30)
Existing home sales	3.05** (1.48)
<i>Investment</i>	
Durable goods	-0.12 (0.85)
Construction spending	-4.77** (2.02)
<i>Output</i>	
GDP	0.02 (1.96)
<i>Government</i>	
Federal budget	1.35 (2.34)
<i>Net exports</i>	
US International trade	-0.32 (0.51)
<i>Prices</i>	
CPI	3.54 (3.29)
<i>Forward-looking</i>	
Consumer confidence	1.53 (1.38)
ISM PMI	6.16 (3.99)
ISM N-Mfg PMI	2.08 (1.68)
N	2367
Adj. R ²	0.140

Note: Model 4 type-specific FPU interaction coefficients δ_j with HC3 (heteroskedasticity-consistent) standard errors in parentheses, as an alternative to the event-time clustered standard errors in Table 7. Coefficient (bps). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

TABLE A2 — MODEL 4 WITH LAGGED FPU (ES)

	δ_j (×FPU)
<i>Real activity</i>	
NFP	8.63** (3.69)
Unemployment rate	-5.41** (2.44)
Retail sales	-4.98*** (1.93)
Industrial production	-0.70 (1.01)
Consumer credit	-0.93* (0.49)
<i>Consumption</i>	
New home sales	0.13 (0.89)
Existing home sales	1.84 (1.53)
<i>Investment</i>	
Durable goods	-0.26 (0.65)
Construction spending	-2.14 (1.79)
<i>Output</i>	
GDP	-1.02 (1.76)
<i>Government</i>	
Federal budget	1.12 (1.46)
<i>Net exports</i>	
US International trade	-0.32 (0.56)
<i>Prices</i>	
CPI	4.57 (3.41)
<i>Forward-looking</i>	
Consumer confidence	0.97 (0.75)
ISM PMI	5.29** (2.60)
ISM N-Mfg PMI	1.52 (1.27)
N	2372
Adj. R ²	0.132

Note: Model 4 estimated with the prior-month FPU value (FPU_{t-1}) in place of the contemporaneous value, addressing within-month look-ahead. δ_j coefficients (bps), event-time clustered standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

TABLE A3 — MODEL 4 ON WINSORIZED SURPRISES (ES)

	δ_j ($\times$ FPU)
<i>Real activity</i>	
NFP	7.52* (4.01)
Unemployment rate	-8.48** (3.77)
Retail sales	-3.80** (1.86)
Industrial production	-1.06 (0.96)
Consumer credit	-1.19** (0.51)
<i>Consumption</i>	
New home sales	-0.22 (1.08)
Existing home sales	2.93** (1.38)
<i>Investment</i>	
Durable goods	-0.46 (0.78)
Construction spending	-6.38*** (2.05)
<i>Output</i>	
GDP	0.00 (2.00)
<i>Government</i>	
Federal budget	1.87 (2.94)
<i>Net exports</i>	
US International trade	-0.55 (0.62)
<i>Prices</i>	
CPI	-0.22 (1.87)
<i>Forward-looking</i>	
Consumer confidence	1.34 (1.31)
ISM PMI	5.79* (3.49)
ISM N-Mfg PMI	2.09 (1.67)
N	2826
Adj. R ²	0.085

*Note: Model 4 type-specific FPU interaction coefficients δ_i estimated on surprises standardized using forecast errors winsorized at the 1st and 99th percentiles, as an alternative treatment of extreme surprises. Coefficient (bps), standard error in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.*

TABLE A4 — ALTERNATIVE EVENT WINDOWS

Window	N	R ²	# sig. (p<.05)
5min [0,+5] MAIN	2826	0.114	5
10min [0,+10]	2826	0.101	3
15min [0,+15]	2826	0.097	4
30min [0,+30]	2826	0.086	4
sym10 [-5,+5]	2826	0.100	2

Note: Main Model 4 (continuous FPU) re-estimated on alternative return windows. N is the number of event-time observations; the final column counts type-specific FPU interactions significant at the 5% level. The [0,+5] window is the main specification.

TABLE A5 — FPU THRESHOLD SENSITIVITY (DUMMY SPECIFICATION)

Threshold	Cutoff	N stressed	% stressed	# sig. (p<.05)
70th pct	151.9	846	29.9	5
75th pct	182.0	707	25.0	3
80th pct	206.7	556	19.7	3
90th pct	306.2	282	10.0	1

Note: Dummy-specification Model 4 re-estimated with the fiscal-stress threshold set at the 70th, 75th, 80th, and 90th percentiles of the monthly FPU distribution. The 75th percentile is the main benchmark. The final column counts type-specific interactions significant at the 5% level.

